

C.WDerive the formula:

$$(i) \bar{x} = A + \frac{1}{N} \sum f_i d_i$$

$$(ii) \bar{x} = A + \frac{h}{N} \sum f_i d_i$$

$$d_i = \frac{x_i - A}{h}$$

$$d_i = x_i - A \text{ (if there is no } h)$$

(i) $\Rightarrow$ we have,

$$d_i = x_i - A$$

$$\Rightarrow f_i d_i = f_i x_i - f_i A$$

$$\cancel{\Rightarrow f_i d_i}$$

Taking summation on both side over i from 1 to n .

$$\sum_{i=1}^n f_i d_i = \sum_{i=1}^n f_i x_i - A \sum_{i=1}^n f_i$$

$$N = \sum f_i$$

$$\Rightarrow \sum_{i=1}^n f_i d_i = \sum_{i=1}^n f_i x_i - AN$$

$$\Rightarrow \frac{\sum_{i=1}^n f_i d_i}{N} = \frac{\sum_{i=1}^n f_i x_i}{N} - A$$

$$\Rightarrow A + \frac{1}{N} \sum_{i=1}^n f_i d_i = \frac{1}{N} \sum_{i=1}^n f_i x_i$$

$$\Rightarrow \bar{x} = A + \frac{1}{N} \sum_{i=1}^n f_i d_i \quad \left[\because \bar{x} = \frac{\sum_{i=1}^n f_i x_i}{N} \right]$$

$$(ii) \Rightarrow$$

we have,

$$d_i = \frac{x_i - A}{h}$$

$$\Rightarrow h d_i = x_i - A$$

$$\Rightarrow h f_i d_i = f_i (x_i - A) = f_i x_i - A f_i$$

Taking summation on both side over i from 1 to n .

$$h \sum_{i=1}^n f_i d_i = \sum_{i=1}^n f_i x_i - A \sum_{i=1}^n f_i$$

$$\Rightarrow h \sum_{i=1}^n f_i d_i = \sum_{i=1}^n f_i x_i - A N \quad \left[N = \sum_{i=1}^n f_i \right]$$

$$\Rightarrow \frac{h}{N} \sum_{i=1}^n f_i d_i = \frac{1}{N} \sum_{i=1}^n f_i x_i - A$$

$$\Rightarrow \frac{h}{N} \sum_{i=1}^n f_i d_i + A = \bar{x} \quad \left[\because \bar{x} = \frac{1}{N} \sum_{i=1}^n f_i x_i \right]$$

$$\Rightarrow \bar{x} = A + \frac{h}{N} \sum_{i=1}^n f_i d_i \quad (\text{shown})$$

[Chapter 1 End]

Extra: • merits & demerits of mean, median, mode.